

Jisang Lee

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Education

Hanbat National University (Advisor: Prof. Haneol Jang)

M.S. Student in Computer Engineering

Daejeon

Mar. 2025 - Present

Fourth semester in progress. **Cumulative GPA 4.5/4.5** through the first three semesters.

Hanbat National University (Advisor: Prof. Haneol Jang)

B.S. in Computer Engineering

Daejeon

Mar. 2019 - Feb. 2025

GPA 4.09/4.5 (Major: 4.28/4.5)

Publications

- [1] **J.S. Lee**, H.W. Seo, E.K. Kim, and H. Jang, "HFGF-DINOv3: High-Frequency Guided Fusion and Phase-Aware Loss for Camouflaged Object Detection", *Pattern Recognition*, 2026. Available online.
- [2] **J.S. Lee**[†], S.Y. Oh[†], C. Park, and H. Jang, "Uncertainty-Guided Structural Routing and Consensus-Aware Re-Ranking for Open-Vocabulary Camouflaged Object Segmentation", *Asian Conference on Computer Vision (ACCV)*, 2026. To appear.
- [3] J.H. Park, **J.S. Lee**, S.M. Pyo, and H. Jang, "Enhanced Copy-Paste Data Augmentation Using SAM for Precise Ship Detection", *Journal of IKEEE*, 2025.
- [4] **J.S. Lee**, H.W. Seo, W.J. Park, H. Choi, K. Kim, E.K. Kim, and H. Jang, "Crop-Paste Data Augmentation Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images", *Journal of IKEEE*, 2025.
- [5] H.W. Seo, **J.S. Lee**, W.J. Park, H. Choi, K. Kim, and H. Jang, "Effective Data Preprocessing and Augmentation Methods of SAR Images for Optical Image Translation", *Journal of Broadcast Engineering*, 2025.
- [6] H.W. Seo, D.S. Kim, S.Y. Oh, **J.S. Lee**, and H. Jang, "Effective Detection of Generated Images Using Frequency Transform", *Journal of KIISE (JOK)*, 2025.
- [7] **J.S. Lee**, H. Kim, J. Koo, H. Choi, and D.Y. Jeong, "Parcel-Based Crop Type Classification in UAV Imagery with SAM for Smallholder Farms", *Korean Journal of Remote Sensing*, 2024.
- [8] **J.S. Lee**, H.W. Seo, C. Park, and H. Jang, "Crop-Paste Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images", *Korea Computer Congress (KCC)*, 2024.
- [9] H.W. Seo, **J.S. Lee**, C. Park, and H. Jang, "Data Preprocessing and Augmentation Methods for SAR to Optical Image Translation", *Korea Computer Congress (KCC)*, 2024.
- [10] D.S. Kim, S.Y. Oh, **J.S. Lee**, and H. Jang, "Effective Data Augmentations for Cross-Domain Fingerprint Recognition", *Journal of Digital Forensics*, 2024.
- [11] H.W. Seo, **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform", *Korean Digital Forensics Society (KDFS) Winter Conference*, 2023.
- [12] **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Lightweight Technique for Forged Fingerprint Detection Using Ensemble Models and Median Filters", *Journal of Digital Forensics*, 2023.
- [13] **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Lightweight Fingerprint Forgery Detection Technique Utilizing Median Filter and Ensemble Model", *Korean Digital Forensics Society (KDFS) Summer Conference*, 2023.
- [14] D.S. Kim, **J.S. Lee**, S.Y. Oh, and H. Jang, "Effective Data Augmentations for Fingerprint Recognition", *Korean Digital Forensics Society (KDFS) Summer Conference*, 2023.
- [15] S.Y. Oh, **J.S. Lee**, D.S. Kim, and H. Jang, "Lightweight Deep Convolutional Neural Networks for Fine-Grained Leaf Classification", *Joint Conference on Communications and Information (JCCI)*, 2023.
- [16] E.J. Jang, **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Development of Optical Character Recognition Model for Document Leak Investigation", *Korean Digital Forensics Society (KDFS) Winter Conference*, 2022.

[†]Equal contribution.

Research Experience

AIM Lab., Hanbat National University

M.S. Researcher (Advisor: Prof. Haneol Jang)

Daejeon, Republic of Korea

Mar. 2025 - Present

- Camouflaged and open-vocabulary segmentation with vision foundation and vision-language models; multispectral image understanding

AIM Lab., Hanbat National University

Undergraduate Researcher (Advisor: Prof. Haneol Jang)

Daejeon, Republic of Korea

May 2022 - Feb. 2025

- Conducted research on computer vision and deep learning

SI Analytics

Research Intern

Daejeon, Republic of Korea

Jul. 2024 - Sep. 2024

- Improved SAM-based parcel segmentation for UAV crop classification; published in KJRS (2024)

Research Projects

Open-Vocabulary Zero-Shot Detection for Military Objects

Collaborative Research with DataMaker

Jan. 2025 - Oct. 2025

- Studied zero-shot military vehicle detection with vision-language models and evaluated recognition of unseen categories

SAR Image Analysis and Semantic Segmentation

Capstone Design with SkyPixel

Jan. 2024 - Dec. 2024

- Studied SAR despeckling, scaling, and generative data augmentation; applied knowledge distillation to segmentation

X-ray Diagnosis of Neonatal Respiratory Distress Syndrome

Chungnam National University Hospital

Jan. 2023 - Jul. 2023

- Developed neonatal X-ray lung segmentation and classification pipelines, with Grad-CAM for interpretation

Awards and Honors

Excellence Award	National Park Satellite Monitoring AI Challenge (2nd/81 teams)	2026
Gold Prize	Capstone Design Exhibition (SkyPixel), Hanbat National University	2024
Prize (3 awards)	Computer Engineering Portfolio Competition, Hanbat National University	2022–2024
Grand Prize	6th AI SPARK Challenge (1st/141 teams); Innopolis Foundation Chairman’s Award	2024
Best Paper Award	KDFS Winter Conference; Society President’s Award	2023
Special Prize	SW-Centered University Joint AI Competition (17th/227 teams); DEEPNOID Award	2023
Best Paper Award	KDFS Summer Conference; Korea Coast Guard Commissioner General Award	2023
5th/542 teams	Fake or Real: AI-Generated Image Detection Challenge, AI Connect	2023
Best Paper Award	KDFS Winter Conference; Society President’s Award	2022
Special Prize	Creative Design Online Competition; Nanum Tech President’s Award	2022
Encouragement Prize	SW-Centered University Joint AI Competition, Finals (8th/77 teams); SW-Centered University Council President’s Award	2022
1st/192 teams	SW-Centered University Joint AI Competition, Preliminaries	2022

Technical Skills

Programming	Python, Java, C, C++
Libraries	PyTorch, NumPy, Pandas, scikit-learn
Languages	Korean (native), English (beginner)
Certifications	Deep Learning Specialization, Coursera (five courses), 2023 Fingerprint Liveness Detection Competition (LivDet), Challenge 1, 2023

Teaching Experiences

Hanbat National University

Teaching Assistant (Prof. Haneol Jang)

2023 - 2026

- Computer Vision (graduate): Fall 2026 (ongoing); Spring 2026
- Computer Vision (4th-year undergraduate): Fall 2025
- Artificial Intelligence and Applications (3rd-year undergraduate): Spring 2025; Spring 2024
- Data Structure (2nd-year undergraduate): Spring 2023
- Hanbat AI Intensive Course: Jan. 2023

Academic Service

IEEE Transactions on Multimedia (TMM)

Reviewer

2026